

Cellular Respiration

How cells convert the chemical energy in glucose into ATP. This handout is intentionally structured to test automatic flashcard generation from definitions, sequences, comparisons, quantities, and cause-and-effect relationships.

Big idea

Cellular respiration transfers energy from glucose to ATP through a series of controlled redox reactions. In aerobic respiration, oxygen is the final electron acceptor. Most ATP is produced by oxidative phosphorylation, not directly by glycolysis or the citric acid cycle.

Learning goals

- Identify the location, inputs, and outputs of each major stage.
- Explain how electron carriers and oxygen support the electron transport chain.
- Compare aerobic respiration with fermentation and predict the effects of pathway disruptions.

Overall chemical equation

Aerobic respiration

$C_6H_{12}O_6 + 6 O_2 \text{ yields } 6 CO_2 + 6 H_2O + \text{about } 30 \text{ to } 32 \text{ ATP}$

Glucose is oxidized, while oxygen is reduced. The equation summarizes many enzyme-controlled reactions rather than one single step.

Core vocabulary

Term	Meaning	Why it matters
ATP	Adenosine triphosphate, the cell's immediate energy currency.	Hydrolysis of its terminal phosphate can power cellular work.
Oxidation	Loss of electrons or hydrogen atoms.	Glucose becomes oxidized as its electrons are removed.
Reduction	Gain of electrons or hydrogen atoms.	Oxygen becomes reduced and forms water.
NADH	A reduced electron carrier made from NAD ⁺ .	Delivers high-energy electrons to the electron transport chain.
FADH ₂	A reduced electron carrier made from FAD.	Also donates electrons to the electron transport chain, but at a lower-energy entry point.
Chemiosmosis	Movement of H ⁺ ions down an electrochemical gradient.	Drives ATP synthase during oxidative phosphorylation.

The four major stages

The totals below are reported per one glucose molecule. Because one glucose produces two pyruvate molecules, pyruvate oxidation and the citric acid cycle occur twice per glucose.

1. Glycolysis

Location: cytosol. **Oxygen:** not directly required.

- One 6-carbon glucose is split into two 3-carbon pyruvate molecules.
- The pathway invests 2 ATP and later produces 4 ATP, for a net gain of 2 ATP.
- Two NAD⁺ molecules are reduced to form 2 NADH.
- ATP is made by substrate-level phosphorylation, meaning an enzyme directly transfers a phosphate group to ADP.

2. Pyruvate oxidation

Location: mitochondrial matrix in eukaryotic cells.

- Each pyruvate loses one carbon as CO₂ and is converted into acetyl-CoA.
- Per glucose, this stage produces 2 acetyl-CoA, 2 CO₂, and 2 NADH.
- No ATP is produced directly during pyruvate oxidation.

3. Citric acid cycle

Location: mitochondrial matrix. Also called the Krebs cycle.

- Acetyl-CoA joins a 4-carbon acceptor to begin a cycle of reactions.
- The 4-carbon acceptor is regenerated, which allows the cycle to continue.
- Two turns per glucose produce a total of 4 CO₂, 6 NADH, 2 FADH₂, and 2 ATP.
- Most of the captured energy leaves this stage in NADH and FADH₂ rather than ATP.

4. Oxidative phosphorylation

Location: inner mitochondrial membrane. Includes the electron transport chain and chemiosmosis.

- NADH and FADH₂ donate high-energy electrons to protein complexes in the membrane.
- Electron energy powers pumps that move H⁺ from the matrix into the intermembrane space.
- The H⁺ gradient stores potential energy, also called the proton-motive force.
- H⁺ flows back into the matrix through ATP synthase, which phosphorylates ADP to make ATP.
- Oxygen accepts electrons at the end of the chain and combines with H⁺ to form water.

Energy accounting and alternatives

Exact ATP yield varies because cells differ in how they shuttle electrons and how much proton leakage occurs across the inner mitochondrial membrane.

Cause and effect: no oxygen

Without oxygen, the electron transport chain stops. NADH can no longer unload its electrons efficiently, the H⁺ gradient collapses, and oxidative phosphorylation stops. A cell may use fermentation to regenerate NAD⁺ so glycolysis can continue, but ATP yield falls sharply.

Approximate yield per glucose

Stage	Direct ATP	NADH	FADH ₂	Carbon product
Glycolysis	2 net	2	0	2 pyruvate
Pyruvate oxidation	0	2	0	2 CO ₂
Citric acid cycle	2	6	2	4 CO ₂
Oxidative phosphorylation	About 26 to 28	Uses carriers	Uses carriers	H ₂ O
Total	About 30 to 32	10 produced	2 produced	6 CO ₂

Aerobic respiration compared with fermentation

Feature	Aerobic respiration	Fermentation
Requires oxygen?	Yes, as the final electron acceptor.	No.
Main locations	Cytosol and mitochondria in eukaryotes.	Cytosol only.
ATP per glucose	About 30 to 32.	2 net ATP from glycolysis.
Purpose after glycolysis	Fully oxidizes carbon and captures much more energy.	Regenerates NAD ⁺ so glycolysis can continue.
Common products	CO ₂ and H ₂ O.	Lactate, or ethanol plus CO ₂ , depending on the organism.

Application and review

Use the cause-and-effect examples to connect pathway structure with cellular outcomes, then check the misconceptions that commonly appear on biology exams.

Applied example: blocked ATP synthase

A drug blocks ATP synthase

H⁺ cannot flow efficiently through ATP synthase, so ATP production by chemiosmosis falls. The H⁺ gradient becomes very steep, which makes further proton pumping difficult. Electron transport slows, NADH accumulates, and oxygen consumption decreases. Glycolysis may still make a small amount of ATP by substrate-level phosphorylation.

Applied example: an uncoupling protein

H⁺ leaks across the inner membrane

An uncoupling protein lets H⁺ return to the matrix without passing through ATP synthase. Electron transport and oxygen consumption can continue or increase, but less ATP is made. The energy in the H⁺ gradient is released mainly as heat.

Common misconceptions

- Plants perform cellular respiration as well as photosynthesis; plant cells also need ATP.
- Breathing and cellular respiration are related but not identical. Breathing exchanges gases, while cellular respiration is a set of chemical reactions inside cells.
- Oxygen does not directly make ATP. It keeps electron flow possible by accepting electrons at the end of the transport chain.
- The citric acid cycle does not produce most ATP directly. It mainly loads electron carriers.
- Fermentation does not add extra ATP beyond glycolysis. Its key role is regenerating NAD⁺.

Quick self-check

You should be able to explain: where glycolysis occurs; why oxygen is essential to the electron transport chain; how NADH differs from ATP; why the citric acid cycle turns twice per glucose; and why fermentation supports glycolysis but yields much less ATP.